KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

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POWER BI PROJECT ON

"FLIPKART SALES INSIGHTS"

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SECTION : CSE-18

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SUBJECT NAME: POWER BI

PROJECT TOPIC:FLIPKART SALES INSIGHTS

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SEMESTER: 6TH

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PROBLEM STATEMENT:-

In today's competitive e-commerce landscape, companies like Flipkart generate massive volumes of transactional data daily. However, raw data alone does not provide actionable insights unless it is properly analyzed and visualized. Flipkart faces challenges in understanding sales performance across different product categories, customer purchasing behavior, and the effectiveness of payment methods.

Despite having data on orders, product categories, pricing, quantities sold, and customer ratings, there is no centralized analytical system to:

- Track overall revenue trends over time
- Identify top-performing and underperforming products
- Understand customer preferences and satisfaction levels
- Analyze the impact of payment methods on sales
- Detect seasonal or time-based sales patterns

Without a comprehensive dashboard, decision-makers struggle to make data-driven strategies for inventory management, pricing optimization, and customer experience improvement.

OBJECTIVE OF THE PROJECT:-

The primary objective of this project is to design and develop an interactive Power BI dashboard that provides comprehensive insights into Flipkart's sales data. The key objectives are:-

- To analyze total sales and quantity sold across different categories
- To identify top-performing and under-performing product categories
- To understand customer behavior through ratings and purchase patterns
- To analyze payment method preferences and their contribution to revenue
- To study sales trends across different time periods
- To enable interactive filtering and drill-down analysis
- To support data-driven decision-making for business growth

DATASET AND METHODOLOGY:-

Dataset:-

File:- "flipkart_sales.xlsx"

ATTRIBUTES	***DETAILS***
Total Records	1000 orders
Features	9 columns

Column Description:-

COLUMN NAME	***DESCRIPTION***
Order ID	Unique order identifier
Product Name	Name of the product
Category	Type of the product
Price (INR)	Cost of the product
Quantity Sold	Number of items sold
Total Sales (INR)	Total sales of the product
Order Date	Date of purchasing the order
Payment Method	Method of payment
Customer Rating	Rating given by the customer

SOLUTIONS AND FEATURES:-

The solution is implemented using Power BI by following structured steps:-

1.) Data Collection and Preparation:-

The dataset includes Order ID, Product Name, Category, Price, Quantity Sold, Total Sales, Order Date, Payment Method, and Customer Rating.
Data cleaning is performed using Power Query by removing null values, correcting data types, and ensuring consistency.

2.) Data Modelling and Calculations:-

Key measures are created using:-

- Total Sales = SUM(Total Sales)
- Total quantity=SUM(Quantity Sold)
- Average Rating=AVERAGE(Customer Rating)

3.) Dashboard Features:-

- **Sales by category:-** Bar chart showing category-wise revenue contribution.
- **Sales by Payment Method:-** Donut chart representing distribution of payment modes.
- **Sales Trend Analysis:**- Line chart displaying sales patterns over time.
- **Detailed Table:-** Shows transaction-level data for deeper insights.
- **Slicers:-** Filters for category and payment method for interactivity.
- **Drill-Down Feature:-** Allows navigation through time hierarchy.
- **UI Design:-** Clean layout with proper alignment and consistent colour theme.

4.) Screenshots of the dashboard:-

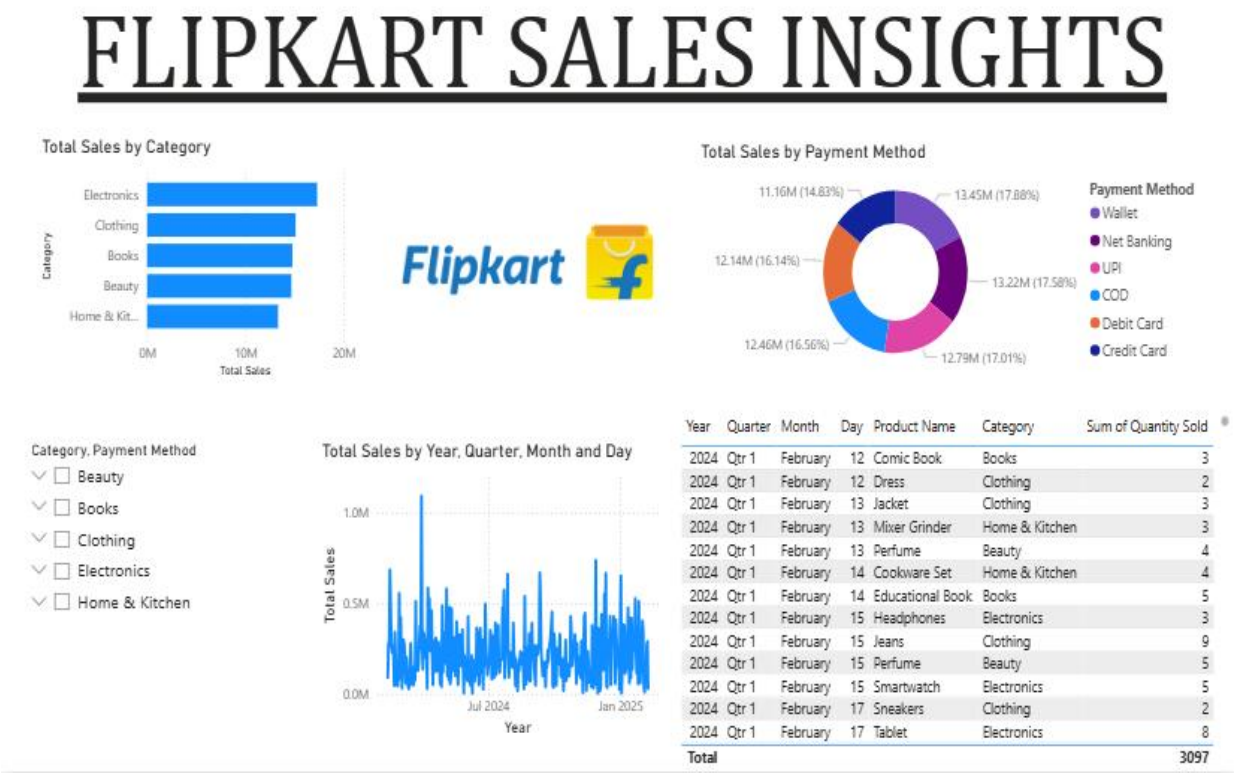

Fig 1:- Preview of the dashboard

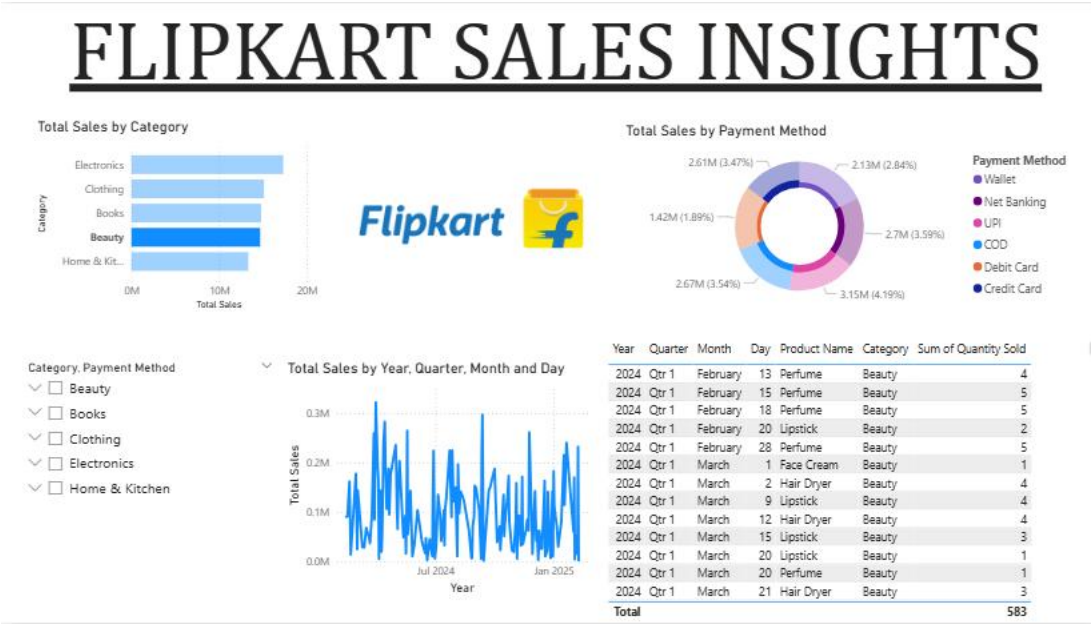

Fig 2: Dashboard when we view total sales by category(for eg:- beauty)

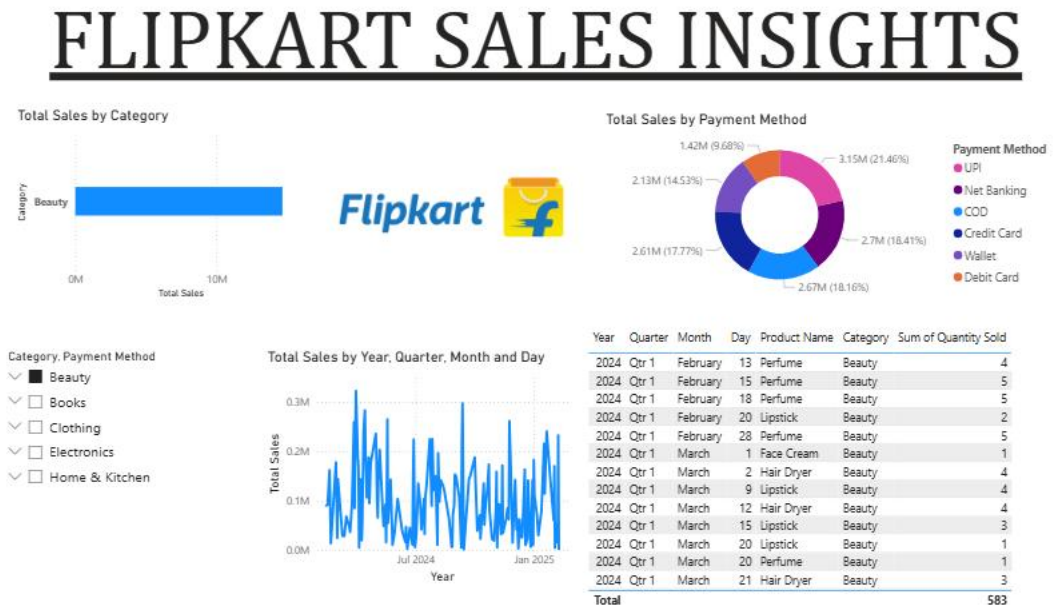

Fig 3:- Slicer used to view payment method by category

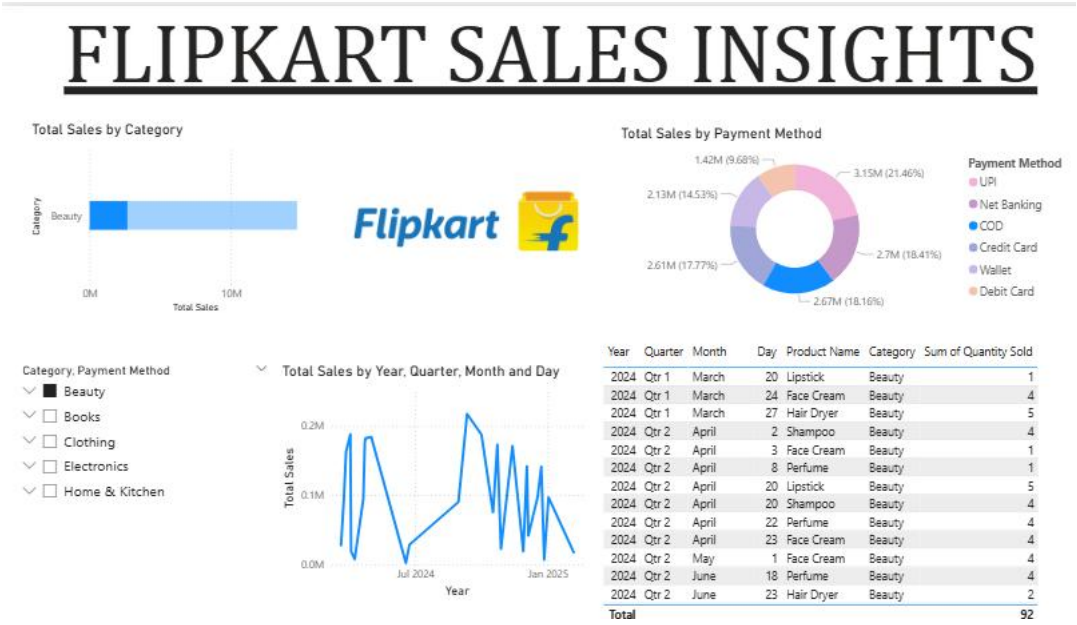

Fig 4:- Changing the donut chart to view total sales by payment method to COD

5.) TECH STACK:-

- **Tool Used:** Microsoft Power BI Desktop
- **Data Source:** Microsoft Excel / CSV
- **Data Transformation:** Power Query Editor
- **Data Analysis:** DAX (Data Analysis Expressions)
- **Visualization Techniques:-**

    1. Bar chart
    2. Donut chart
    3. Line chart
    4. Table
    5. Slicers

6.) UNIQUE POINTS OF THE PROJECT:-

- Interactive dashboard with slicers
- Drill-down analysis for time-based insights
- Clear visualization of business-critical metrics
- Combination of multiple charts for better insights
- User-friendly and professional layout
- Helps simulate real-world e-commerce analytics

7.) KEY INSIGHTS:-

- Electronics and Clothing contribute the highest sales
- Digital payments (UPI, Cards) dominate transactions
- Sales show fluctuations across months (seasonal trends)
- Customer ratings help identify product quality trends

8.) FUTURE IMPROVEMENTS:-

- Add **KPI cards** (Total Sales, Orders, Avg Rating)
- Integrate **real-time data** using APIs
- Implement **forecasting models** for sales prediction
- Add **customer segmentation analysis**
- Include **profit and cost analysis**
- Publish dashboard using Power BI Service for online access

9.) CONCLUSION:-

This project demonstrates how raw sales data can be transformed into actionable insights using Power BI. The dashboard helps in analyzing sales performance, customer behavior, and operational trends, enabling Flipkart to make data-driven decisions.